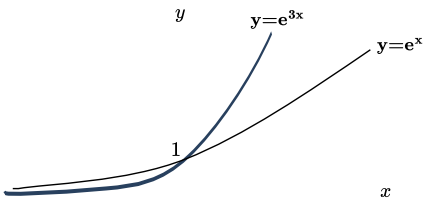
1.25 Graph Sketching: e and Natural Logarithms

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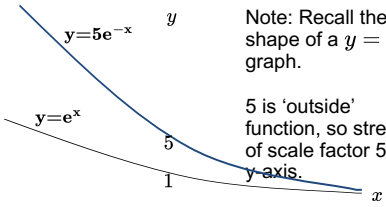
More Graph Transformations

11

Sketch $y = e^{3x}$



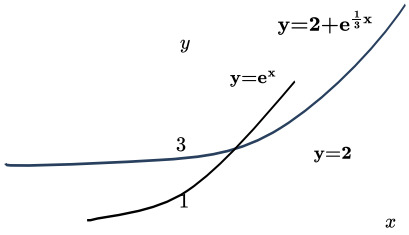
Sketch $y = 5e^{-x}$



Note: Recall the shape of a $y = a^{-x}$ graph.

5 is 'outside' function, so stretch of scale factor 5 on y-axis.

Sketch $y = 2 + e^{\frac{1}{3}x}$



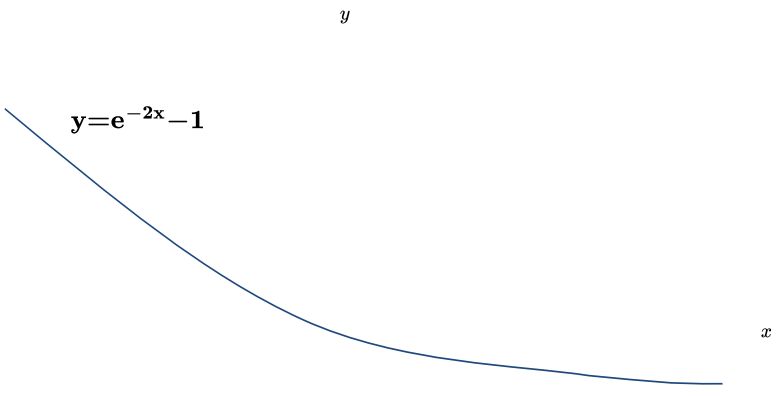
Important Note: Because the original asymptote was $y = 0$, it is now $y = 2$ and you must indicate this along with its equation.

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Test Your Understanding

12

Sketch $y = e^{-2x} - 1$



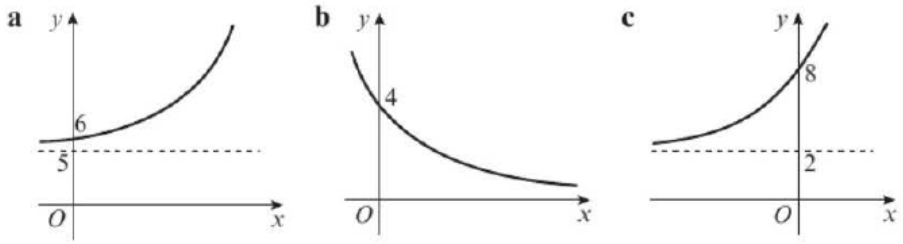
1.25 Graph Sketching: e and Natural Logarithms

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Exercise 14B (Pages 316-317)

13

4 Each of the sketch graphs below is of the form $y = Ae^{bx} + C$, where A, b and C are constants. Find the values of A and C for each graph, and state whether b is positive or negative.



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Exercise 14B (Pages 316-317)

14

6 Differentiate the following with respect to x.

a e^{6x} \qquad **b** $e^{-x/3}$ \qquad **c** $7e^{2x}$

d $5e^{0.4x}$ \qquad **e** $e^{3x} + 2e^{x}$ \qquad **f** $e^{x}(e^{x} + 1)$

1.25 Graph Sketching: e and Natural Logarithms

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Exercise 14B (Pages 316-317)

15

7 Find the gradient of the curve with equation $y = e^{3x}$ at the point where

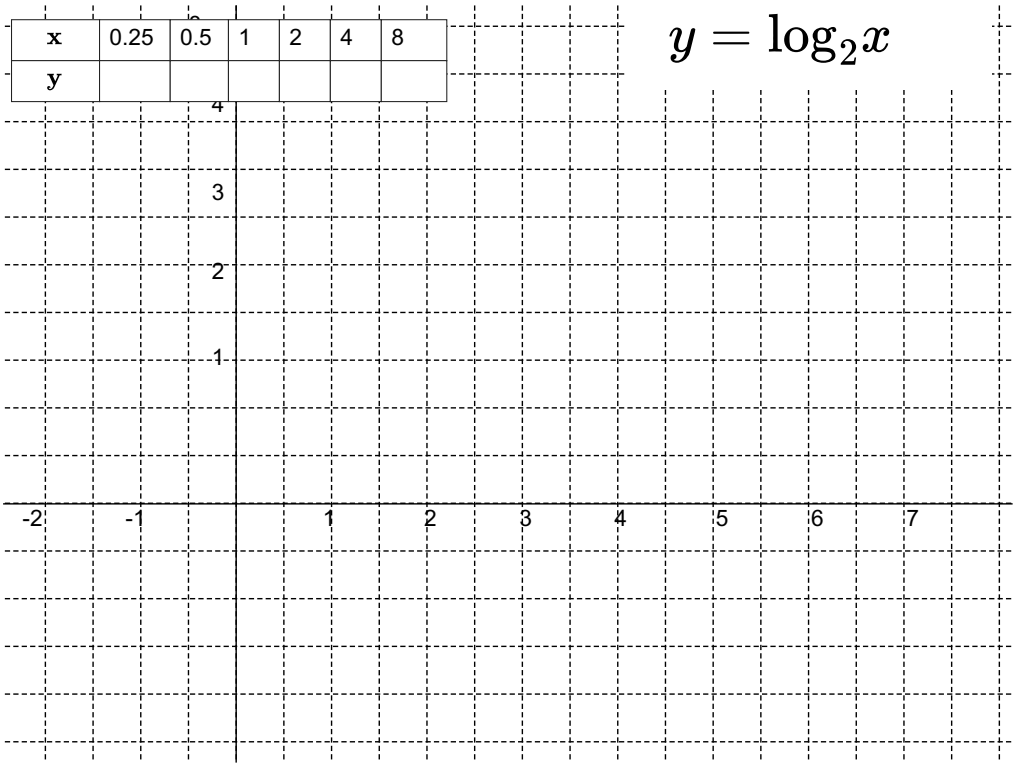
a $x = 2$ **b** $x = 0$ **c** $x = -0.5$

8 The function f is defined as $f(x) = e^{0.2x}$, $x \in \mathbb{R}$. Show that the tangent to the curve at the point (5, e) goes through the origin.

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x	0.25	0.5	1	2	4	8
y						

$y = \log_2 x$



1.25 Graph Sketching: e and Natural Logarithms

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Natural Logarithms

We have previously seen that $y = \log_a x$ is the inverse function of $y = a^x$.
We also saw that e^x is "**the**" exponential function.
The inverse of e^x is $\log_e x$, but because of its special importance, it has its own function name!

The inverse of $y = e^x$ is $y = \ln x$

$\ln e^x =$

$e^{\ln x} =$

Solve $e^x = 5$

"ln both sides". On the LHS it cancels out the "e to the power of"

Solve $2 \ln x + 1 = 5$

Do "e to the power of" each side. On the LHS it cancels out the ln.